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Digital Forensics Court Case Report

Barakah Pay: A Trust on Trial

1. Introduction

This report presents an analysis of the digital forensics courtroom case titled *Barakah Pay: A Trust on Trial*. The project was designed to simulate a real-world cybercrime trial involving financial fraud within a fintech company operating under Shariah-compliant financial principles.

The objective of this assignment was to understand the role of digital forensics in criminal investigations, examine how digital evidence is presented in court, and demonstrate the legal and ethical responsibilities of technology professionals. Through role-play and structured courtroom procedures, students developed technical, legal, analytical, and communication skills.

2. Case Background

The case revolves around BarakahPay, a fintech company that promised ethical and transparent profit-sharing in accordance with Islamic finance principles. The accused, Abdi Rahman, the Chief Technology Officer (CTO), secretly modified a smart contract to divert 0.031% of profits into a private digital wallet.

Although each transaction appeared insignificant, over six months the accumulated amount reached 6.4 million dirhams. The fraud was discovered during a Shariah compliance audit and later confirmed through a digital forensic investigation.

The charges against the defendant included:

- Unauthorized modification of financial systems
- Financial fraud
- Breach of fiduciary trust
- Ethical misconduct

3. Roles and Responsibilities

The courtroom simulation included the following key roles:

- Judge (Hon. Justice Faruk) – Oversaw the trial, ruled on objections, and delivered the final verdict.
- Prosecutor (Atty. Parvej Rafi) – Presented evidence to prove guilt beyond reasonable doubt.
- Defense Attorney (Atty. Hutayfa) – Challenged the prosecution's claims and attempted to create reasonable doubt.
- Digital Forensics Investigator (Mujeeb) – Analyzed blockchain transactions, deployment logs, and digital credentials.
- Shariah Advisory Board Member (Khadija) – Testified on ethical and compliance violations.
- Defendant (Abdi Rahman) – CTO accused of intentionally deploying malicious code.
- Journalist (Jainab) – Represented public interest and societal impact.

Each role contributed to demonstrating how technical and legal professionals interact in cybercrime proceedings.

4. Digital Forensics Investigation

The digital forensic investigation was a critical element of the case. The investigator applied structured forensic methodologies including:

4.1 Evidence Collection

- Blockchain transaction analysis
- Smart contract code review
- Deployment log verification
- Wallet tracing

4.2 Evidence Preservation

- Maintaining chain of custody

- Securing server logs
- Verifying cryptographic integrity

4.3 Evidence Analysis

The investigation revealed:

- Repeated micro-transactions redirecting funds
- A total loss of 6.4 million dirhams
- The hidden function embedded within the smart contract
- Deployment credentials linked directly to the defendant
- No signs of external hacking or unauthorized access

The forensic findings confirmed that the fraud was deliberate and technically sophisticated. The small transaction design demonstrated intent to avoid detection.

5. Legal Analysis of Digital Evidence

For digital evidence to be admissible in court, it must meet standards of:

- Authenticity
- Reliability
- Integrity
- Proper chain of custody

In this case:

- Logs confirmed valid credentials were used.
- Wallet recovery phrases were linked to the defendant.
- No malware or intrusion indicators were found.

The prosecution argued that the consistency of the evidence proved intentional misconduct.

The defense attempted to challenge the evidence by suggesting:

- Possibility of hacking
- Credential compromise
- Lack of direct eyewitness proof

However, the digital forensic investigator testified that no signs of account takeover were detected, strengthening the prosecution's case.

6. Ethical and Shariah Compliance Issues

Beyond technical fraud, the case involved ethical violations. The Shariah Advisory Board member testified that any undisclosed profit-sharing modification constitutes deception.

Key ethical principles violated:

- Transparency
- Trust
- Accountability
- Consent in financial agreements

The case highlighted how technology professionals hold positions of trust and how misuse of system privileges can cause large-scale harm.

7. Courtroom Proceedings

The courtroom followed professional legal structure:

Opening Statements

The prosecution framed the case around "trust," arguing that the defendant engineered a system to steal gradually.
The defense emphasized intent and human error rather than criminal design.

Witness Examination

- The Shariah board member confirmed no approval was granted.
- The digital forensic investigator provided technical proof linking the accused to the wallet and smart contract deployment.
- Under cross-examination, alternative theories such as hacking were considered but unsupported.

Defendant Testimony

The defendant admitted:

- Understanding what the function would do
- Knowing customer funds were being redirected
- Believing he would not be caught

This admission significantly strengthened the prosecution's argument.

8. Verdict

After reviewing the evidence and testimonies, the judge found Abdi Rahman guilty on all charges.

The court concluded:

- The fraud was intentional.
- Trust was deliberately violated.
- Financial harm was real and measurable.

The final message emphasized that while technology systems function correctly, human ethics determine their integrity.

9. Learning Outcomes

This project achieved several educational objectives:

9.1 Technical Skills

- Understanding blockchain forensics
- Smart contract analysis
- Log examination and digital tracing

9.2 Legal Knowledge

- Admissibility of digital evidence
- Burden of proof
- Cross-examination procedures

9.3 Critical Thinking

- Evaluating strengths and weaknesses of digital evidence
- Analyzing intent vs. negligence

9.4 Communication Skills

- Explaining complex technical concepts in simple language
- Public speaking in courtroom simulation

9.5 Ethical Awareness

- Responsibility of IT professionals
- Importance of compliance and governance
- Impact of financial fraud on society

10. Reflection and Evaluation

The courtroom simulation effectively demonstrated how digital forensics bridges technology and law. The case showed that cybercrime does not always involve external hackers; sometimes the threat comes from trusted insiders with system-level access.

The strongest aspect of the case was the structured forensic evidence, which clearly linked the accused to the fraudulent wallet. The defense strategy highlighted the importance of challenging digital evidence, reinforcing the principle of “beyond reasonable doubt.”

If improved further, additional visual aids such as transaction timelines or blockchain flow diagrams could enhance clarity.

Overall, the project successfully illustrated the importance of digital forensic investigation in protecting financial systems and maintaining public trust.

11. Conclusion

The *Barakah Pay: A Trust on Trial* case demonstrates how minor technical manipulations can evolve into major financial crimes. It reinforces that digital forensic science is essential in uncovering hidden cyber fraud and ensuring accountability. The trial concluded with a powerful message

Technology does not commit crimes people do. Digital forensics ensures that even the smallest digital footprint can reveal the truth.